CBSE – CLASS XII PRACTICE QUESTION PAPER

Subject: MATHEMATICS

Chapter: Vector Algebra

Time Allowed: 1 Hour Maximum Marks: 30

General Instructions:

1. All questions are compulsory.
2. The paper has 10 questions in three sections: Section A (Q1–4, 2 marks each), Section B (Q5–8, 3 marks each) and Section C (Q9–10, 5 marks each).
3. Show all steps in numerical problems with proper units. Draw neat, labelled diagrams wherever asked.
4. Vectors are written in bold type. **i**, **j**, **k** denote unit vectors along the x, y and z axes.

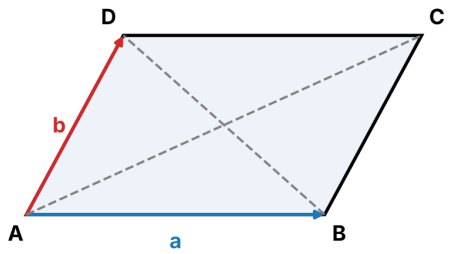
SECTION A (Q1–Q4) – 2 marks each

Q1. Find the unit vector in the direction of the vector $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. **[2]**

Q2. Find the projection of the vector $\mathbf{a} = 2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}$ on the vector $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$. **[2]**

Q3. If **a** and **b** are two unit vectors and θ is the angle between them, such that **a** + **b** is also a unit vector, find θ. **[2]**

Q4. In the figure, ABCD is a parallelogram with AB = **a** and AD = **b**. Express the vectors AC and BD in terms of **a** and **b**, and hence show that AC + BD = 2**b**. **[2]**



SECTION B (Q5–Q8) – 3 marks each

Q5. Find the area of the parallelogram whose adjacent sides are $\mathbf{a} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$. **[3]**

Q6. Show that the points A(1, –2, –8), B(5, 0, –2) and C(11, 3, 7) are collinear, and find the ratio in which B divides AC. **[3]**

Q7. If **a**, **b**, **c** are three vectors such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, $|\mathbf{a}| = 3$, $|\mathbf{b}| = 5$ and $|\mathbf{c}| = 7$, find the angle between **a** and **b**. **[3]**

Q8. Find the position vector of the point R which divides the line segment joining P($\mathbf{i} + 2\mathbf{j} - \mathbf{k}$) and Q($-\mathbf{i} + \mathbf{j} + \mathbf{k}$) in the ratio 2 : 1 (i) internally, and (ii) externally. **[3]**

SECTION C (Q9–Q10) – 5 marks each

Q9. (a) If $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$, find a unit vector perpendicular to both **a** + **b** and **a** – **b**. (b) Hence find the area of the parallelogram whose diagonals are **a** + **b** and **a** – **b**. **[5]**

Q10. The position vectors of the vertices of △ABC are A: $2\mathbf{i} - \mathbf{j} + \mathbf{k}$, B: $\mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$, C: $3\mathbf{i} - 4\mathbf{j} - 4\mathbf{k}$. (a) Show that the triangle is right-angled. (b) Find its area. **[5]**